

CADMIUM**ICSC: 0020 (April 2005)****CAS #: 7440-43-9****UN #: 2570****EC Number: 231-152-8**

	ACUTE HAZARDS	PREVENTION	FIRE FIGHTING
FIRE & EXPLOSION	May ignite spontaneously on contact with air. Gives off irritating or toxic fumes (or gases) in a fire. Finely dispersed particles form explosive mixtures in air.	NO open flames, NO sparks and NO smoking. NO contact with heat or acids. Closed system, dust explosion-proof electrical equipment and lighting. Prevent deposition of dust.	Use dry sand. Use special powder. NO other agents.

PREVENT DISPERSION OF DUST! AVOID ALL CONTACT! IN ALL CASES CONSULT A DOCTOR!

	SYMPTOMS	PREVENTION	FIRST AID
Inhalation	Cough. Sore throat.	Use local exhaust or breathing protection.	Fresh air, rest. Refer for medical attention.
Skin		Protective gloves.	Remove contaminated clothes. Rinse and then wash skin with water and soap.
Eyes	Redness. Pain.	Wear safety goggles or eye protection in combination with breathing protection.	First rinse with plenty of water for several minutes (remove contact lenses if easily possible), then refer for medical attention.
Ingestion	Abdominal pain. Diarrhoea. Headache. Nausea. Vomiting.	Do not eat, drink, or smoke during work.	Rest. Refer for medical attention .

SPILLAGE DISPOSAL	CLASSIFICATION & LABELLING
Evacuate danger area! Personal protection: chemical protection suit including self-contained breathing apparatus. Remove all ignition sources. Sweep spilled substance into covered containers. Carefully collect remainder. Then store and dispose of according to local regulations.	According to UN GHS Criteria Transportation UN Classification UN Hazard Class: 6.1
STORAGE	
Fireproof. Dry. Keep under inert gas. Separated from ignition sources, oxidants, acids and food and feedstuffs.	
PACKAGING	
Airtight. Unbreakable packaging. Put breakable packaging into closed unbreakable container. Do not transport with food and feedstuffs.	

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CADMIUM**ICSC: 0020****PHYSICAL & CHEMICAL INFORMATION****Physical State; Appearance**

SOFT BLUE-WHITE METAL LUMPS OR GREY POWDER.
MALLEABLE. TURNS BRITTLE ON EXPOSURE TO 80°C. TARNISHES ON EXPOSURE TO MOIST AIR.

Physical dangers

Dust explosion possible if in powder or granular form, mixed with air.

Chemical dangers

Reacts with acids. This produces flammable/explosive gas (hydrogen - see ICSC 0001). The dust reacts with oxidants, hydrogen azide, zinc, selenium and tellurium. This generates fire and explosion hazard.

Formula: Cd

Atomic mass: 112.4

Boiling point: 765°C

Melting point: 321°C

Density: 8.6 g/cm³

Solubility in water: none

Auto-ignition temperature: 250°C (cadmium metal dust)

EXPOSURE & HEALTH EFFECTS**Routes of exposure**

The substance can be absorbed into the body by inhalation of its aerosol and by ingestion.

Effects of short-term exposure

The fume is irritating to the respiratory tract. Inhalation of fumes may cause lung oedema. See Notes. Inhalation of fumes may cause metal fume fever. The effects may be delayed. Medical observation is indicated.

Inhalation risk

A harmful concentration of airborne particles can be reached quickly when dispersed, especially if powdered.

Effects of long-term or repeated exposure

Repeated or prolonged inhalation of dust particles may cause effects on the lungs. The substance may have effects on the kidneys. This may result in kidney impairment. This substance is carcinogenic to humans.

OCCUPATIONAL EXPOSURE LIMITS

TLV: 0.01 mg/m³, as TWA; A2 (suspected human carcinogen); BEI issued.

MAK: (including its inorganic compounds, inhalable fraction): skin absorption (H); carcinogen category: 1; germ cell mutagen group: 3A.

EU-OEL: (inhalable fraction): 0.001 mg/m³ as TWA; (see Notes)

ENVIRONMENT**NOTES**

Reacts violently with fire extinguishing agents such as water, foam, carbon dioxide and halons.

Depending on the degree of exposure, periodic medical examination is suggested.

The symptoms of lung oedema often do not become manifest until a few hours have passed and they are aggravated by physical effort. Rest and medical observation are therefore essential.

Do NOT take working clothes home.

UN numbers and packing group will vary according to the physical form of the substance.

An EU-OEL of 0.004 mg/m³ is allowed until 11 July 2027.

ADDITIONAL INFORMATION**EC Classification**

Symbol: T+, N; R: 45-26-48/23/25-62-63-68-50/53; S: 53-45-60-61; Note: E

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